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AQA A Level Further Mathematics (7367)

Topic **DA: Graphs**

Sub topic **DA1: Terminology for Graphs**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **DA: Graphs**

Sub topic **DA1: Terminology for Graphs**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

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Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

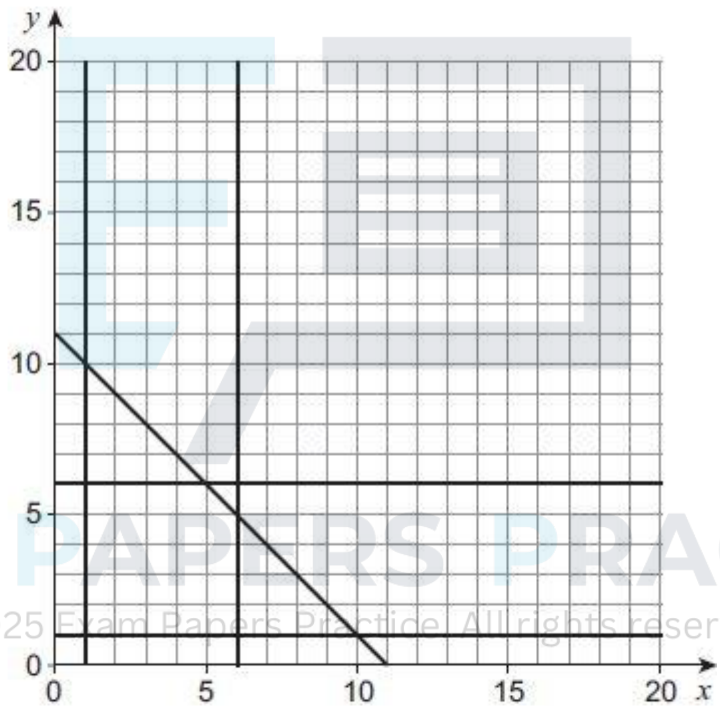
Q1.

A linear programming problem has the constraints

$$1 \le x \le 6$$
$$1 \le y \le 6$$
$$y \ge x$$
$$x + y \le 11$$

(a) (i) Complete **Figure 1** to identify the feasible region for the problem.

Figure 1



(2)

(ii) Determine the maximum value of $5x + 4y$ subject to the constraints.

(2)

(b) The simple-connected graph G has seven vertices.

The vertices of G have degree 1, 2, 3, v, w, x and y

(i) Explain why $x \ge 1$ and $y \ge 1$

(1)

(ii) Explain why $x \le 6$ and $y \le 6$

(1)

(iii) Explain why $x + y \le 11$.

(2)

(iv) State an additional constraint that applies to the values of x and y in this context.

(1)

(c) The graph G also has eight edges. The inequalities used in part (a)(i) apply to the graph G.

(i) Given that $v + w = 4$, find all the feasible values of x and y.

(3)

(ii) It is also given that the graph G is semi-Eulerian.

On **Figure 2**, draw G.

Figure 2

(2)
(Total 14 marks)

Q2.

A graph has 5 vertices and 6 edges.

Find the sum of the degrees of the vertices.

Circle your answer.

10 11 12 15

(Total 1 mark)

Q3.

(a) Draw a bipartite graph to represent the following adjacency matrix.

	1	2	3	4	5
A	0	0	0	0	1
B	0	0	0	1	1
C	0	1	0	1	1
D	0	0	0	1	1
E	1	1	1	0	1

(2)

(b) If A, B, C, D and E represent five people and 1, 2, 3, 4 and 5 represent five tasks to which they are to be assigned, explain why a complete matching is impossible.

(2)

(Total 4 marks)

Q4.

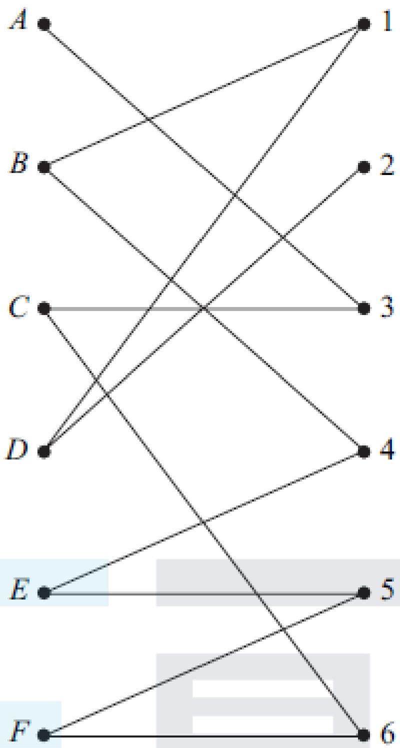
Draw a bipartite graph representing the following adjacency matrix.

	1	2	3	4	5	6
A	1	1	0	0	1	1
B	0	1	0	0	1	1
C	1	0	0	0	1	1
D	0	1	1	1	0	1
E	0	0	0	0	1	1
F	0	0	0	0	0	1

(Total 2 marks)

Q5.

Six people, A, B, C, D, E and F , are to be allocated to six tasks, 1, 2, 3, 4, 5 and 6.
The following bipartite graph shows the tasks that each of the people is able to undertake.



Represent this information in an adjacency matrix.

(Total 2 marks)

Q6.

The complete graph K_n ($n > 1$) has every one of its n vertices connected to each of the other vertices by a single edge.

(a) Draw the complete graph K_4.

(1)

(b) (i) Find the total number of edges for the graph K_8.

(ii) Give a reason why K_8 is not Eulerian.

(2)

(c) For the graph K_n, state in terms of n:

(i) the total number of edges;

(ii) the number of edges in a minimum spanning tree;

(iii) the condition for K_n to be Eulerian;

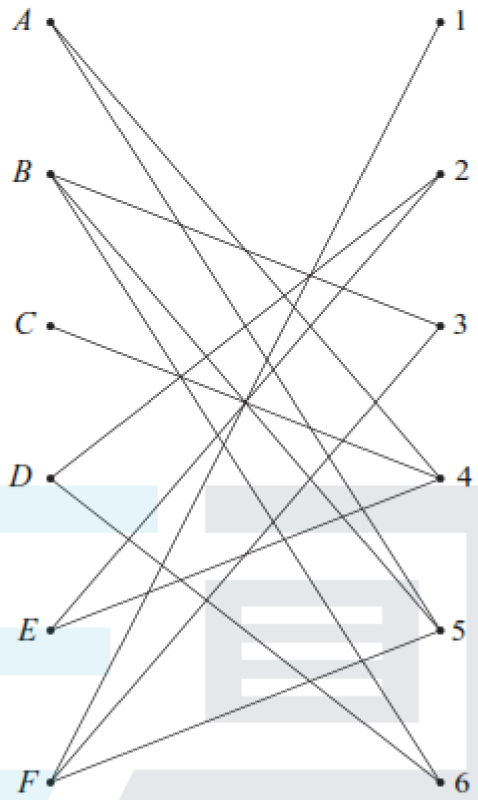
(iv) the condition for the number of edges of a Hamiltonian cycle to be equal to the number of edges of an Eulerian cycle.

(4)

(Total 7 marks)

Q7.

Six people, A, B, C, D, E and $F,$ are to be allocated to six tasks, 1, 2, 3, 4, 5 and 6. The following bipartite graph shows the tasks that each of the people is able to undertake.



Represent this information in an adjacency matrix.

(Total 2 marks)

Q8.

A simple connected graph has six vertices.

(a) One vertex has degree x. State the greatest and least possible values of x.

(2)

(b) The six vertices have degrees

$$x - 2,\ x - 2,\ x,\ 2x - 4,\ 2x - 4,\ 4x - 12$$

Find the value of x, justifying your answer.

(2)

(Total 4 marks)

Q9.

A connected graph G has five vertices and has eight edges with lengths 8, 10, 10, 11, 13, 17, 17 and 18.

(a) Find the minimum length of a minimum spanning tree for G.

(2)

(b) Find the maximum length of a minimum spanning tree for G.

(2)

(c) Draw a sketch to show a possible graph G when the length of the minimum spanning tree is 53.

(3)
(Total 7 marks)

Q10.

Angelo is visiting six famous places in Palermo: A, B, C, D, E and F. He intends to travel from one place to the next until he has visited all of the places before returning to his starting place. Due to the traffic system, the time taken to travel between two places may be different dependent on the direction travelled.

The table shows the times, in minutes, taken to travel between the six places.

From \ To	A	B	C	D	E	F
A	–	25	20	20	27	25
B	15	–	10	11	15	30
C	5	30	–	15	20	19
D	20	25	15	–	25	10
E	10	20	7	15	–	15
F	25	35	29	20	30	–

(a) Give an example of a Hamiltonian cycle in this context.

(2)

(b) (i) Show that, if the nearest neighbour algorithm starting from F is used, the total travelling time for Angelo would be 95 minutes.

(3)

(ii) Explain why your answer to part (b)(i) is an upper bound for the minimum travelling time for Angelo.

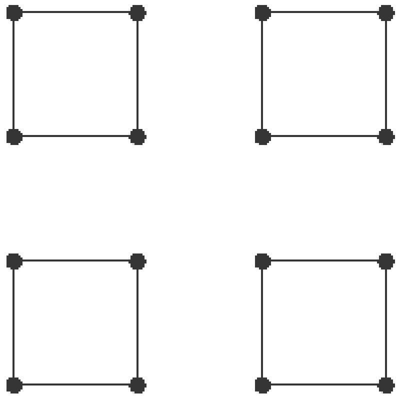
(2)

(c) Angelo starts from F and visits E next. He also visits B before he visits D. Find an improved upper bound for Angelo's total travelling time.

(3)
(Total 10 marks)

Q11.

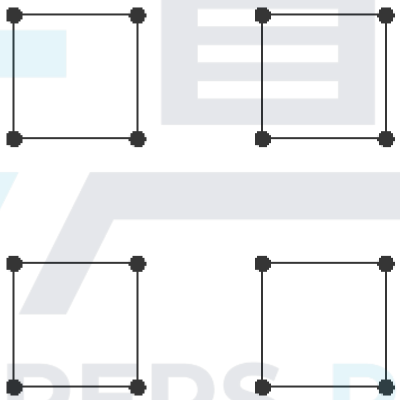
(a) The diagram shows a graph with 16 vertices and 16 edges.



(i) On **Figure 1** below, add the minimum number of edges to make a connected graph.

Figure 1

Connected Graph

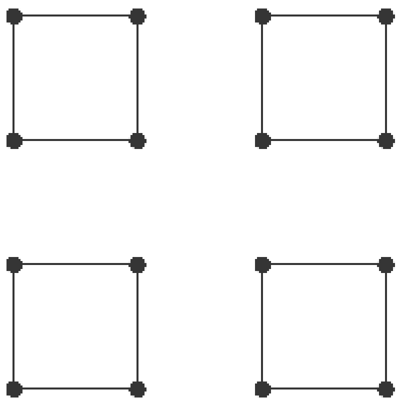


(1)

(ii) On **Figure 2** below, add the minimum number of edges to make the graph Hamiltonian.

Figure 2

Hamiltonian Graph

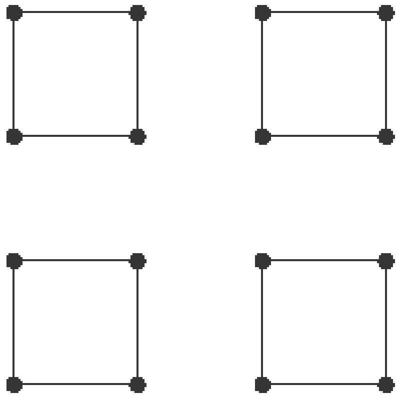


(2)

(iii) On **Figure 3** below, add the minimum number of edges to make the graph Eulerian.

Figure 3

Eulerian Graph



(2)

(b) A complete graph has n vertices and is Eulerian.

(i) State the condition that n must satisfy.

(1)

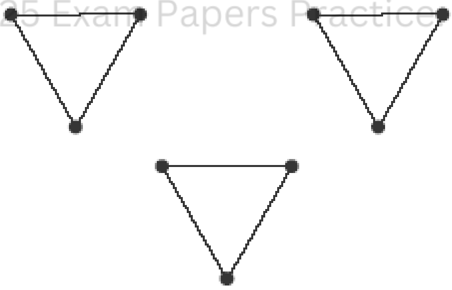
(ii) The number of edges in a Hamiltonian cycle for the graph is the same as the number of edges in an Eulerian trail. State the value of n.

(2)

(Total 8 marks)

Q12.

(a) The diagram shows a graph **G** with 9 vertices and 9 edges.



(i) State the minimum number of edges that need to be added to **G** to make a connected graph. Draw an example of such a graph.

(2)

(ii) State the minimum number of edges that need to be added to **G** to make the graph Hamiltonian. Draw an example of such a graph.

(2)

(iii) State the minimum number of edges that need to be added to **G** to make the graph Eulerian. Draw an example of such a graph.

(2)

(b) A complete graph has n vertices and is Eulerian.

(i) State the condition that n must satisfy.

(1)

(ii) In addition, the number of edges in a Hamiltonian cycle for the graph is the same as the number of edges in an Eulerian trail. State the value of n.

(1)
(Total 8 marks)

Q13.

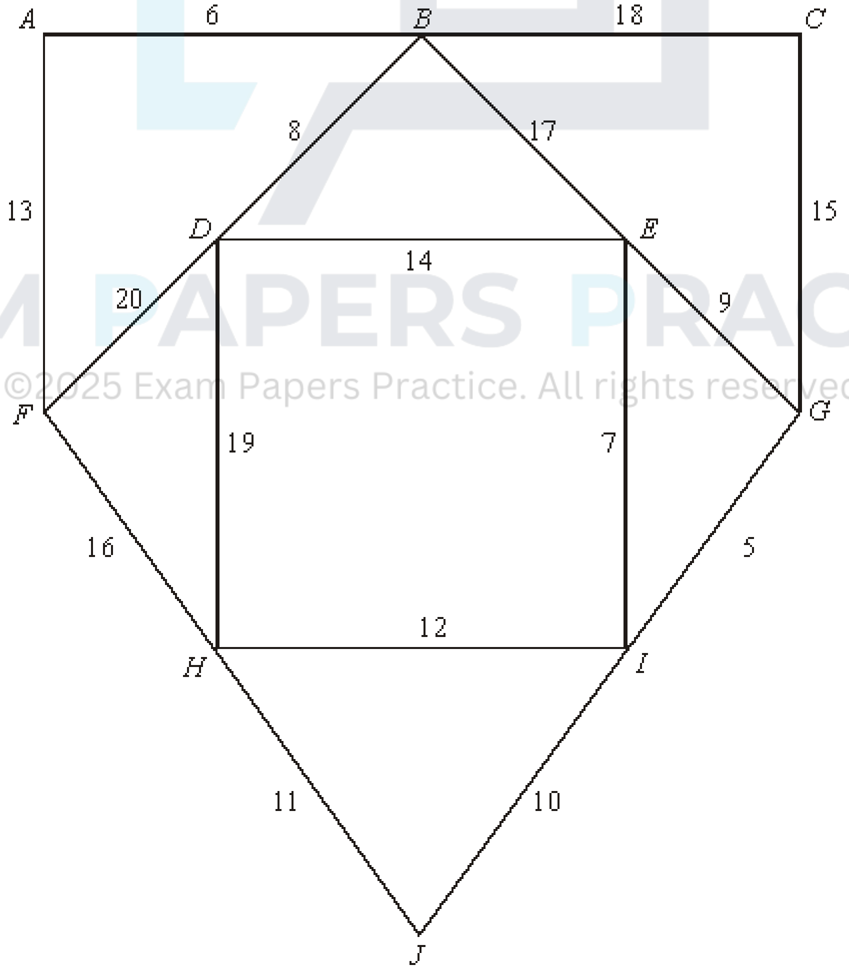
(a) (i) State the number of edges in a minimum spanning tree of a network with 10 vertices.

(1)

(ii) State the number of edges in a minimum spanning tree of a network with n vertices.

(1)

(b) The following network has 10 vertices: A, B, ..., J. The numbers on each edge represent the distances, in miles, between pairs of vertices.



(i) Use Kruskal's algorithm to find the minimum spanning tree for the network.

(5)

(ii) State the length of your spanning tree.

(1)

(iii) Draw your spanning tree.

(2)
(Total 10 marks)

Q14.

A connected graph **G** has m vertices and n edges.

(a) (i) Write down the number of edges in a minimum spanning tree of **G**.

(1)

(ii) Hence write down an inequality relating m and n.

(2)

(b) The graph **G** contains a Hamiltonian cycle. Write down the number of edges in this cycle.

(1)

(c) In the case where **G** is Eulerian, draw a graph of **G** for which m = 6 and n = 12.

(2)
(Total 6 marks)

Q15.

(a) In the complete graph K_7, every one of the 7 vertices is connected to each of the other 6 vertices by a single edge.

Find or write down:

(i) the number of edges in the graph;

(1)

(ii) the number of edges in a minimum spanning tree;

(1)

(iii) the number of edges in a Hamiltonian cycle.

(1)

(b) (i) Explain why the graph K_7 is Eulerian.

(1)

(ii) Write down the condition for K_n to be Eulerian.

(1)

(c) A connected graph has 6 vertices and 10 edges.

Draw an example of such a graph which is Eulerian.

(2)
(Total 7 marks)

Q16.

A connected graph has five vertices and has arc lengths of

4, 7, 7, 7, 8, 8, 9 and 12 units.

(a) State the minimum length of a minimum spanning tree for any such graph.

(1)

(b) State the minimum length of a Hamiltonian cycle for any such graph.

(1)

(c) State the minimum length of an Eulerian cycle for any such graph.

(1)

(d) In the case when the length of its minimum spanning tree is 26 units, draw a sketch to show a possible graph.

(3)

(Total 6 marks)

Q17.

(a) Draw the graph K_4.

(2)

(b) (i) Find the total number of edges in K_6.

(1)

(ii) State the number of edges in a minimum spanning tree of K_6.

(1)

(iii) Give a reason why K_6 is **not** Eulerian.

(1)

(c) For the graph K_n, state in terms of n:

(i) the total number of edges;

(1)

(ii) the number of edges in a minimum spanning tree;

(1)

(iii) the condition for K_n to be Eulerian.

(1)

(Total 8 marks)

Q18.

A graph G has five vertices.

(a) (i) Given that G is connected, state the number of arcs on a minimum spanning tree.

(1)

(ii) Given that G is Eulerian with all vertices having the same degree d, state the minimum value of d.

(1)

(iii) Given that G has a Hamiltonian cycle, state the number of edges in such a cycle.

(1)

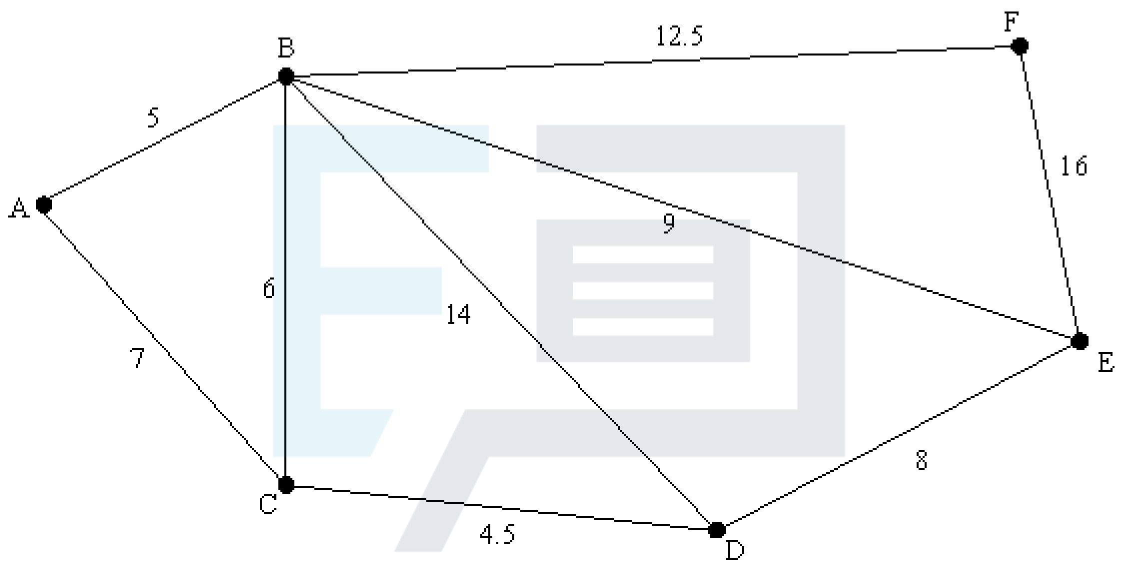
(b) Given that G has all the properties mentioned in part (a), draw a possible graph G.

(1)
(Total 4 marks)

Q19.

A local council is responsible for gritting roads.

(a) The following diagram shows the lengths of roads, in miles, that have to be gritted.



The gritter is based at A and must travel along all the roads, at least once, before returning *to* A.

(i) Explain why it is **not** possible to start from A and, by travelling along each road only once, return to A.

(1)

(ii) Find an optimal ‘Chinese postman’ route around the network, starting and finishing at A. State the length of your route.

(6)

(b) (i) The connected graph of the roads in the area run by another council has six odd vertices. Find the number of ways of pairing these odd vertices.

(1)

(ii) For a connected graph with n odd vertices, find an expression for the number of ways of pairing these vertices.

(2)
(Total 10 marks)

Q20.

(a) A connected graph has four vertices. State the number of edges in the graph’s minimum spanning tree

(1)

(b) A graph has n vertices. The graph is complete, i.e. each vertex is joined to every other vertex by exactly one edge.

(i) State the number of edges in the graph's minimum spanning tree

(1)

(ii) Determine the number of Hamiltonian cycles in the graph.

(2)

(c) A connected graph has four vertices and has arc lengths of

4, 4.5, 5, 6.5, 7, 8 and 9 units.

The length of its minimum spanning tree is 17 units. Draw a sketch to show a possible graph.

(3)

(Total 7 marks)

Q21.

(a) (i) Draw the edges of a graph with the following vertex to vertex table.

	A	B	C	D
A	0	1	2	1
B	1	0	2	1
C	2	2	0	1
D	1	1	1	0

(2)

(ii) Explain why this graph does **not** have an Eulerian cycle.

(1)

(b) A vertex to vertex table is symmetrical about a leading diagonal consisting only of zeros. State the connection between the sum of all the numbers in the table and the number of edges in the corresponding graph.

(1)

(c) State the circumstances in which a graph of a vertex to vertex table is **not** symmetrical about its main diagonal.

(1)

(Total 5 marks)

Q22.

A connected graph G has five vertices. The lengths of the edges are 7, 7, 7, 9, 11, 12, 13, 15 and 16 units, respectively.

(a) A graph is said to be *fully connected* if every vertex is joined at least once to every other vertex. Explain why graph G cannot be fully connected.

(1)

(b) A connected graph G is to be drawn with 5 vertices with lengths of edges as given above

(i) Calculate the least possible length of a minimum spanning tree of graph G.

(1)

(ii) Explain why your answer to part (b)(i) might not apply to graph G.

(1)

(iii) Sketch an example of graph G which has a minimum spanning tree of length 34 units.

(3)

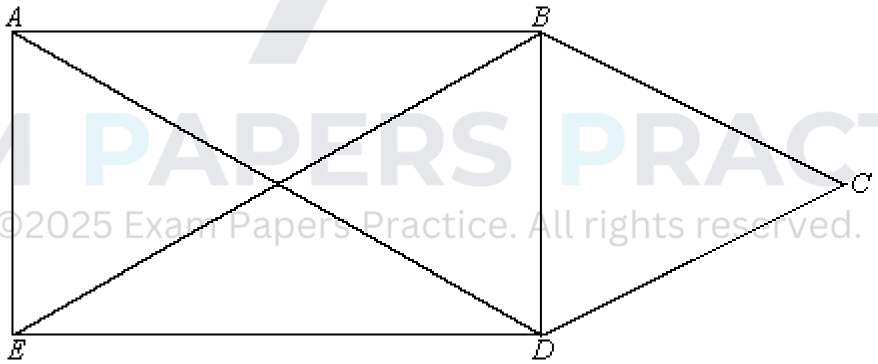
(Total 6 marks)

Q23.

(a) Explain what is meant by an Eulerian cycle.

(1)

(b) (i) Explain why the following graph cannot have an Eulerian cycle.



(1)

(ii) The graph can be drawn without taking your pen off the paper and without retracing any line. State the vertices from which you could start the drawing.

(1)

(iii) Write down a path that would enable you to draw the graph without taking your pen off the paper and without retracing any line.

(1)

(Total 4 marks)

Q24.

(a) Explain what is meant by a Hamiltonian cycle.

(2)

(b) Determine how many different Hamiltonian cycles there are in a fully connected graph with four vertices.

(2)

(c) Write down the number of different Hamiltonian cycles there are in a fully connected graph with n vertices.

(2)

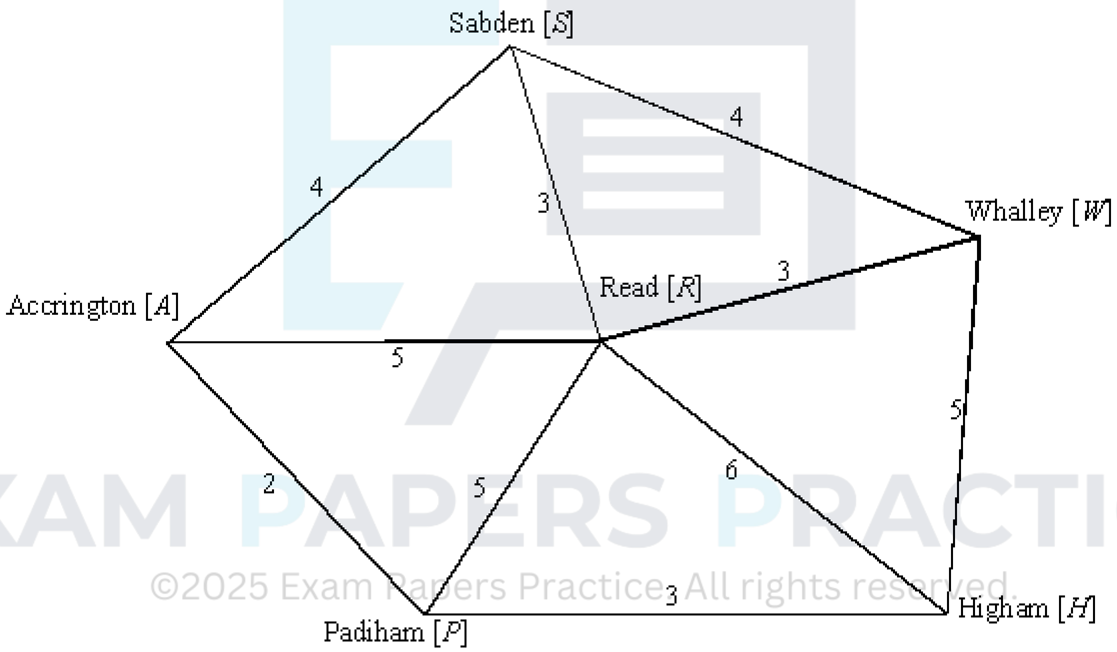
(Total 6 marks)

Q25.

(a) For a connected graph with n vertices, state the number of edges in a minimum spanning tree.

(1)

(b) A cable company has laid cables to Accrington and is now considering extending the network to neighbouring villages. The distances, in miles, between the villages are shown.



Using Prim's algorithm, showing a sketch at each stage, obtain the minimum spanning tree for the cable company.

(5)

(c) State the length of your minimum spanning tree.

(1)

(Total 7 marks)

Q26.

(a) Draw a simple connected graph with 9 vertices all having degree 2.

(2)

(b) A simple connected graph has 9 vertices, all having the same degree d.

(i) State the possible values of d.

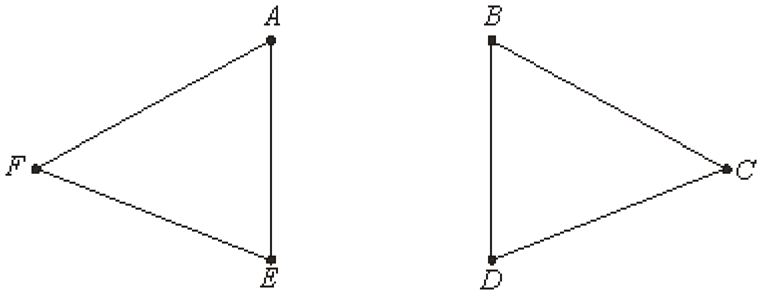
(2)

(ii) For **each** of these values of d, state the number of edges of the graph.

(3)
(Total 7 marks)

Q27.

The graph **G** has six vertices $A - F$ and is illustrated below.



(a) What is the minimum number of edges which must be added to **G** in order to make the graph connected?

(1)

(b) What is the minimum number of edges which must be added to **G** in order to make the graph Hamiltonian? State a suitable set of edges.

(2)

(c) (i) State the minimum number of edges which must be added to **G** in order to create a semi-Eulerian graph. Draw such a graph and write down a trail of this graph which uses all its edges.

(3)

(ii) Show that by adding two edges between the same pair of vertices you can create an Eulerian graph.

(1)

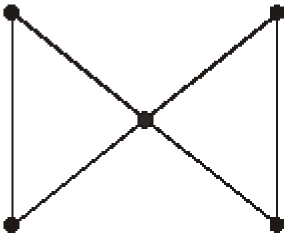
(iii) State the minimum number of edges which must be added to **G** in order to create a **simple** graph which is Eulerian. Draw such a graph.

(3)
(Total 10 marks)

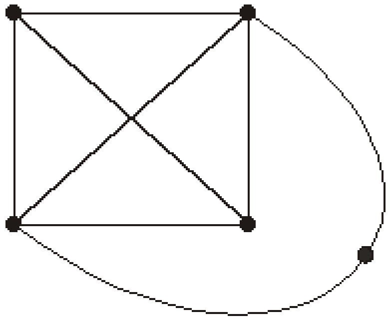
Q28.

The four graphs **E**, **F**, **G** and **H** are illustrated below. They each have five vertices.

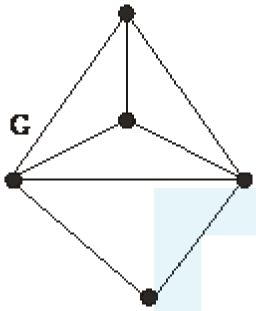
E



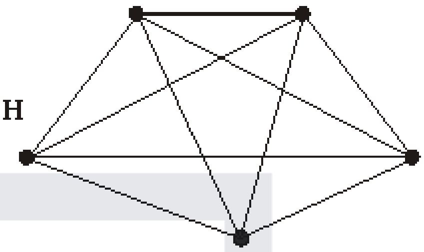
F



G



H



(a) State which, if any, of the four graphs are Eulerian.

(2)

(b) State which, if any, of the four graphs are Hamiltonian.

(2)

(c) State which, if any, of the four graphs are planar.

(2)

(d) State which pairs, if any, of the four graphs are isomorphic.

(2)

(Total 8 marks)

Q29.

(a) A simple graph has five vertices and their degrees are d, $d + 1$, $d + 1$, $d + 2$ and $d + 3$.

(i) By considering the sum of the degrees, show that d is odd.

(2)

(ii) Deduce that $d = 1$ and draw a graph with vertices having the given degrees.

(3)

(b) A simple graph has 10 vertices.

(i) Explain why the degree of each vertex must be in the set

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

(1)

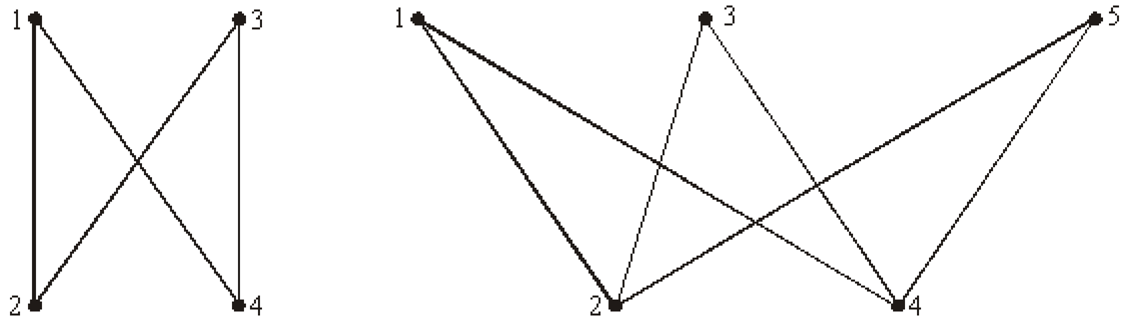
(ii) Show that the degrees of the vertices cannot all be different.

(2)

(Total 8 marks)

Q30.

G$_n$ is the graph with n vertices labelled 1, 2, 3, . . ., n and with two vertices joined by an edge if the sum of their labels is odd. So, for example, **G**$_4$ and **G**$_5$ are as drawn below.



$\mathbf{G_4}$ $\mathbf{G_5}$

(a) Draw **G**$_6$.

(2)

(b) (i) Give an example of a Hamiltonian cycle in **G**$_6$.

(2)

(ii) For what values of n is **G**$_n$ Hamiltonian?

(2)

(c) If n is even what, in terms of n, is the degree of each vertex of **G**$_n$?

(1)

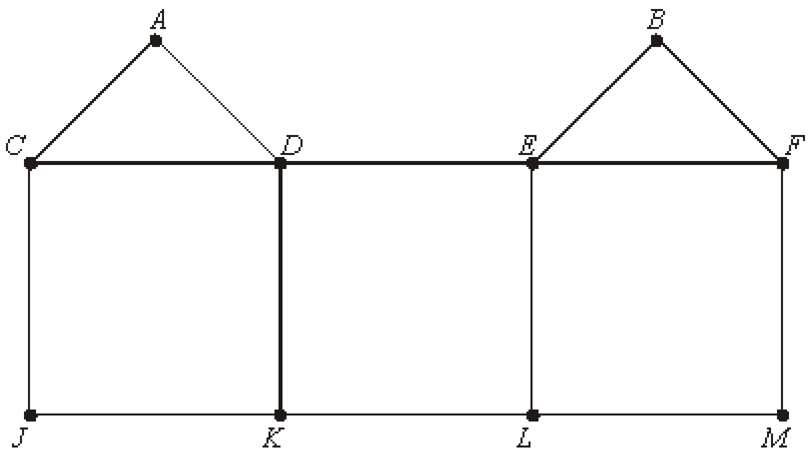
(d) For what values of n is **G**$_n$ Eulerian? Justify your answer.

(3)

(Total 10 marks)

Q31.

G is the graph illustrated.



(a) Find the Hamiltonian cycle in **G** which begins CA ....

(2)

(b) Is **G** Eulerian, semi-Eulerian or neither? Give a reason for your answer.

(2)

(c) One edge is removed from **G** to make a semi-Eulerian graph.

(i) State which edge is removed.

(1)

(ii) Explain why the resulting graph is not Hamiltonian.

(2)
(Total 7 marks)

Q32.

A simple graph has six vertices and their degrees are d, $d + 3$, $d + 3$, $d + 3$, $d + 3$ and d + 4.

(i) Explain why d must be either 0 or 1.

(1)

(ii) For which of these values of d will the graph be semi-Eulerian?
Give a reason.

(2)

(iii) Show that the graph will never be Hamiltonian.

(2)
(Total 5 marks)

Q33.

A graph **G** has seven vertices labelled 51, 52, 53, 61, 62, 63 and 64. Two vertices are joined by an edge if their two labels use four different digits. (So, for example, 51 is joined to 62, but 51 is not joined to 52 or to 61.)

(a) Draw the graph **G**.

(2)

(b) Explain how you know that **G** is not Eulerian or semi-Eulerian.

(1)

(c) Find an example of a path in **G** from 51 to 61 which uses exactly three edges. How many paths are there from 51 to 61 which use an even number of edges? Explain briefly why each cycle in **G** uses an even number of edges.

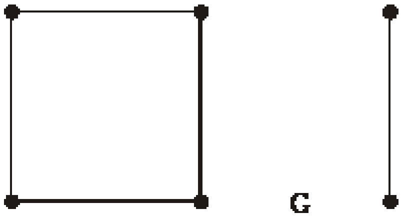
(4)

(d) Deduce that **G** is not Hamiltonian.

(2)
(Total 9 marks)

Q34.

G is the graph illustrated with 6 vertices and 5 edges:



(a) What is the smallest number of edges which must be added to **G** in order to make a connected graph? Illustrate one such graph,

(1)

(b) What is the smallest number of edges which must be added to **G** in order to make a Hamiltonian graph? Illustrate one such graph, clearly pointing out a Hamiltonian cycle.

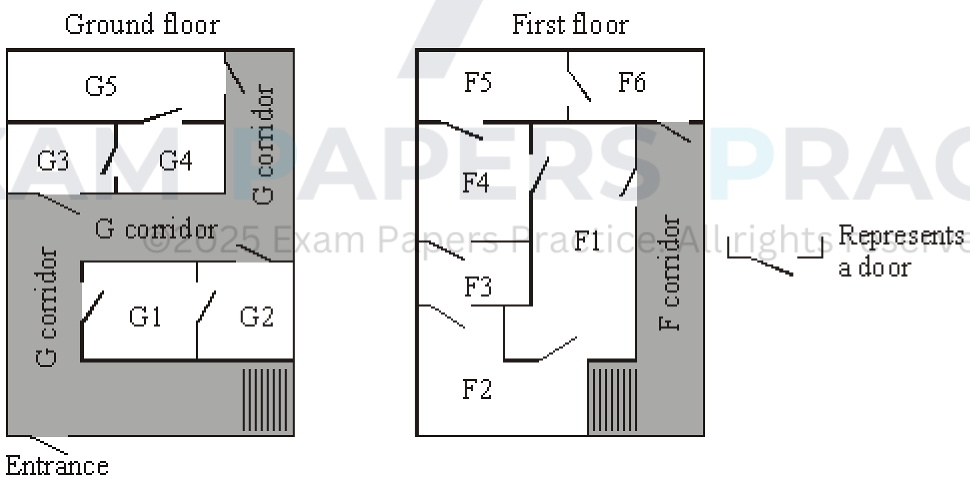
(2)

(c) What is the smallest number of edges which must be added to **G** in order to make an Eulerian graph? Illustrate one such graph.

(2)
(Total 5 marks)

Q35.

The diagram below shows plans of the two floors of a building.



(a) Draw a graph to represent the ground floor with a vertex for each of the five rooms and one vertex for the corridor, and with edges showing which rooms or the corridor are directly connected by a door. Ignore the stairs.

(2)

(b) Similarly, draw a graph for the first floor,

(2)

(c) Show that one of your graphs is Eulerian and that the other is semi-Eulerian, and give an interpretation of this for each floor.

(3)
(Total 7 marks)

Q36.

EXAM PAPERS PRACTICE

A graph has eight vertices labelled 1, 2, 3, 4, 5, 6, 7, 8. Two vertices are joined by an edge if their numbers differ by one or two. For example, 2 and 3 are joined, 4 and 6 are joined, but 5 and 8 are not joined.

(a) Draw a picture of the graph.

(3)

(b) Explain how you know that the graph is semi-Eulerian.
Give an example of an Eulerian trail in the graph.

(3)

(c) Give an example of a Hamiltonian cycle in the graph.

(2)

(d) List all paths of length four from vertex 1 to vertex 8.

(3)
(Total 11 marks)

Q37.

A simple graph has vertices A, B, C, D, E, F. The degree of vertex A is m and the degree of each of the other vertices is $2m$, where m is a positive integer.

(a) By considering the sum of the degrees show that m is even, and deduce that $m = 2$.

(2)

(b) Draw such a graph.

(3)

(c) Calculate the number of graphs which satisfy the given description.

(2)
(Total 7 marks)

Q38.

A university mathematics lecturer is organising her six students into pairs to work on a mathematical modelling exercise. She would like to arrange it so that no students who have previously worked together are doing so on this exercise. So far

a has worked with b, d and e,
b has worked with a, d and f,
c has worked with d, e and f,
d has worked with a, b and c,
e has worked with a, c and f,
and f has worked with b, c and e.

(a) Draw a graph, G, to represent this information.

(3)

(b) Draw K_6, the complete graph on six vertices.

(1)

(c) Draw a graph on six vertices which consists of all those edges which are in K_6 but **not** in G. Say how this will help the lecturer.

(3)
(Total 7 marks)

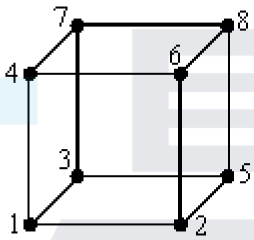
Q39.

A pumping room contains three large electrically powered pumps, pump A, pump B and pump C. They are all currently pumping. Any pump can be switched off if it is on (or on if it is off), but only one at a time may be switched to avoid damaging power surges.

(a) Draw a graph in which the vertices represent the possible states (on or off) of each pump, and in which the edges connect pairs of vertices which differ by just one switching. The state when all three are on is represented by the vertex (A, B, C), and this vertex is connected by an edge to the vertex (~A, B, C), and to some other vertices.

(3)

(b) Show that your graph in part (a) is isomorphic to the graph shown below.



(2)

(c) Give the number of paths with three edges connecting (A, B, C) to (~A, ~B, ~C).

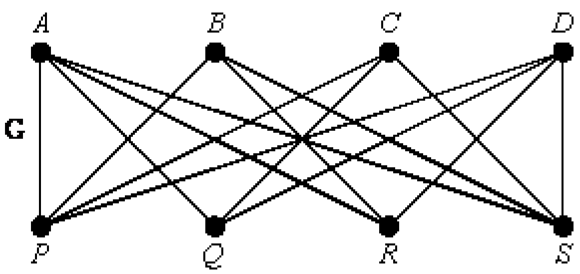
(2)

(d) Give the number of paths which have 5 edges, none repeated, connecting (A, B, C) to (~A, ~B, ~C).

(2)
(Total 9 marks)

Q40.

The graph **G** shown is the graph $K_{4,4}$.



(a) Explain what is meant by a Hamiltonian cycle, and give an example from **G.**

(3)

(b) How many different Hamiltonian cycles are there in **G?**
(Do not count a cycle taken in reverse order as different from the original.)

(2)

(c) **G** is non-planar.

Explain what this means.
Use Kuratowski's theorem to prove that **G** is non-planar.

(2)
(Total 7 marks)